

COURSE TITLE : RUBBER TECHNOLOGY LAB
COURSE CODE : 3097
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER: 60/3
CREDITS : 2

TIME SCHEDULE

Module	Topics	Period
1	Mastication and mixing of NR	16
2	Compounding of synthetic rubbers	16
3	Compression moulding, extrusion and calendaring processes	16
4	Test specimen preparation	12
TOTAL		60

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To do mastication and mixing of NR
2	2	To compound synthetic rubbers
3	3	To conduct compression moulding, extrusion and calendaring processes
4	4	To prepare test specimens for hardness, tensile properties, tear strength, compression set.

LIST OF EXPERIMENTS

MODULE I

- 1.1. Identify the processing machineries mixing mill screw press and hydraulic press
- 1.2. Perform mastication of NR in a lab size mixing mill
- 1.3. Carry out blending of NR with SBR
- 1.4. Carry out blending of NR with BR
- 1.5. Preparation of NR gum compound
- 1.6. Preparation of NR masterbatch with carbon black
- 1.7. Preparation of NR masterbatch with clay or whiting
- 1.8.1.8. Perform compound preparation of NR with black and nonblack filler.

MODULE II

- 2.1. Perform warming of SR in a lab size mixing mill
- 2.2. Preparation of SR gum compound
- 2.3. Perform compound preparation of SBR, NBR and EPDM
- 2.4. Preparation of SBR masterbatch with carbon black
- 2.5. Perform compound preparation of SBR with black and nonblack filler.

MODULE III

- 3.1. Carry out compression moulding with NR and SBR compound.
- 3.2. Carry out extrusion of NR and SBR compound.
- 3.3. Carry out calendaring with NR and SBR compound

MODULE IV

- 4.1. Prepare test specimens for hardness.
- 4.2. Prepare test specimens for Tensile properties .
- 4.3. Prepare test specimens for tear strength.
- 4.4. Prepare test specimens for compression set.